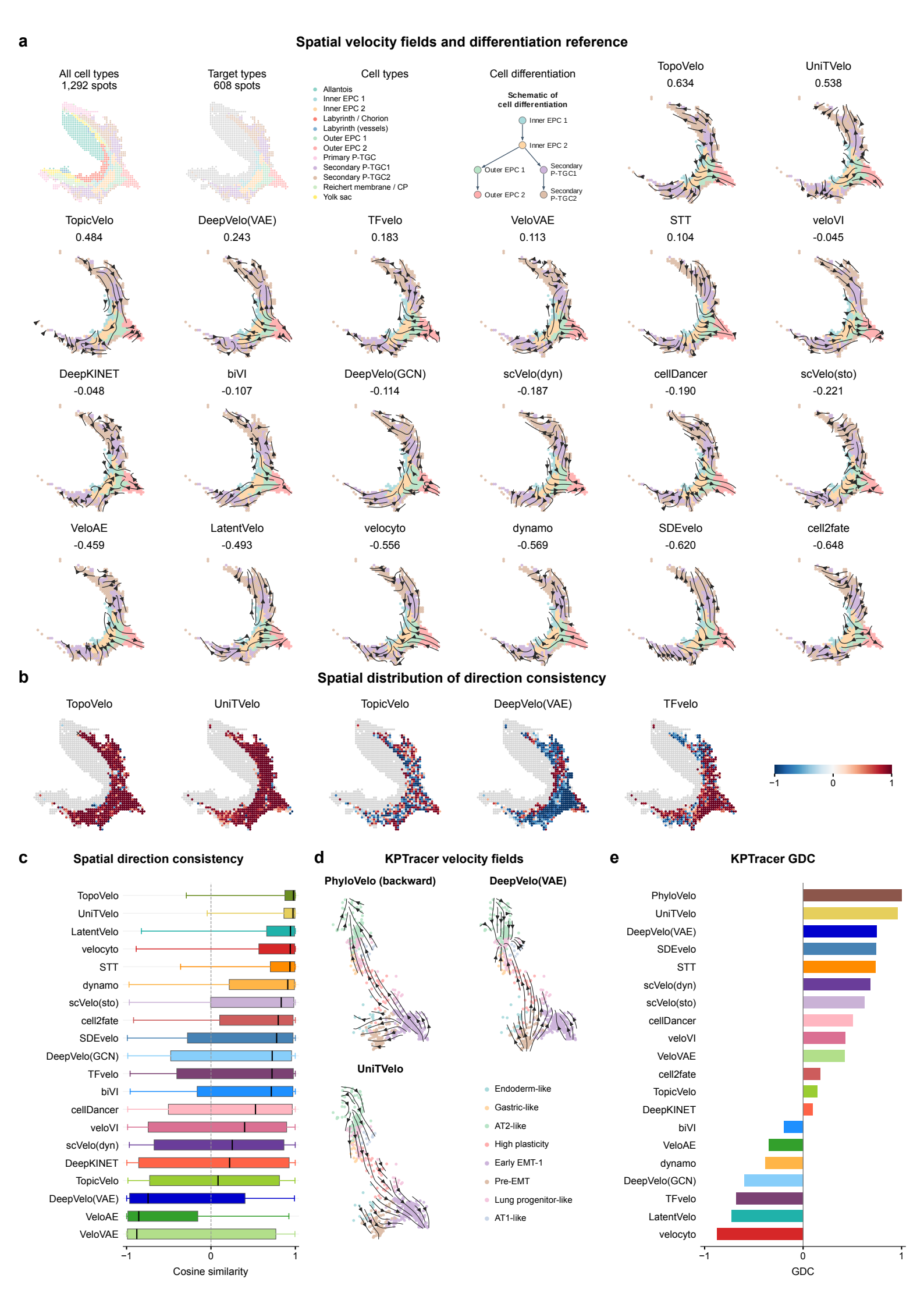

a
Spatial velocity fields and differentiation reference
All cell types
1,292 spots
Target types
608 spots
Cell types
Allantois
Inner EPC 1
Inner EPC 2
Labyrinth / Chorion
Labyrinth (vessels)
Outer EPC 1
Outer EPC 2
Primary P-TGC
Secondary P-TGC1
Secondary P-TGC2
Reichert membrane / CP
Yolk sac
Cell differentiation
Schematic of cell differentiation
Inner EPC 1
Inner EPC 2
Outer EPC 1
Secondary P-TGC1
Outer EPC 2
Secondary P-TGC2
TopoVelo
0.634
UniTVelo
0.538
TopicVelo
0.484
DeepVelo(VAE)
0.243
TFvelo
0.183
VeloVAE
0.113
STT
0.104
veloVI
-0.045
DeepKINET
-0.048
biVI
-0.107
DeepVelo(GCN)
-0.114
scVelo(dyn)
-0.187
cellDancer
-0.190
scVelo(sto)
-0.221
VeloAE
-0.459
LatentVelo
-0.493
velocyto
-0.556
dynamo
-0.569
SDEvelo
-0.620
cell2fate
-0.648
b
Spatial distribution of direction consistency
TopoVelo
UniTVelo
TopicVelo
DeepVelo(VAE)
TFvelo
−1
0
1
c
Spatial direction consistency
TopoVelo
UniTVelo
LatentVelo
velocyto
STT
dynamo
scVelo(sto)
cell2fate
SDEvelo
DeepVelo(GCN)
TFvelo
biVI
cellDancer
veloVI
scVelo(dyn)
DeepKINET
TopicVelo
DeepVelo(VAE)
VeloAE
VeloVAE
−1
0
1
Cosine similarity
d
KPTracer velocity fields
PhyloVelo (backward)
DeepVelo(VAE)
UniTVelo
Endoderm-like
Gastric-like
AT2-like
High plasticity
Early EMT-1
Pre-EMT
Lung progenitor-like
AT1-like
e
KPTracer GDC
PhyloVelo
UniTVelo
DeepVelo(VAE)
SDEvelo
STT
scVelo(dyn)
scVelo(sto)
cellDancer
veloVI
VeloVAE
cell2fate
TopicVelo
DeepKINET
biVI
VeloAE
dynamo
DeepVelo(GCN)
TFvelo
LatentVelo
velocyto
−1
0
1
GDC

Fig. 5 | Spatial context and reference-based evaluation of velocity directions. a, Spatial fields for 20 methods on the 608-spot MPSTA ST_small subset, ordered by descending GDC from left to right and top to bottom. The first four tiles show all 1,292 spots, the target spots, cell-type colors and five schematic reference transitions. TopoVelo GDC is 0.634 (607 nonzero gene-velocity predictions and one zero vector); TopicVelo GDC is 0.484 and uses native-transition displacement T @ Ms - Ms. **b,** Paired spatial-direction cosine maps for the five highest-GDC methods in a: TopoVelo, UniTVelo, TopicVelo, DeepVelo(VAE) and TFvelo. Colors indicate agreement between saved full-context and restricted-context directions; black crosses indicate undefined pairs. **c,** The same spatial-direction consistency across all 20 methods, ordered by median cosine. Boxes show the median and interquartile range; whiskers show the 5th–95th percentiles. TopoVelo has 606 valid pairs and other methods have 608. These within-dataset comparisons measure saved-workflow sensitivity, not an isolated causal effect of tissue context. **d,** KPTracer fields for PhyloVelo (backward), UniTVelo and DeepVelo(VAE), using the same 754 cells, fixed UMAP, equal aspect and shared streamline parameters. PhyloVelo retains its native backward arrows; evaluation uses the negated vectors to align the forward direction. **e,** KPTracer scores sorted by displayed value. The reference denominator is 706 cells in six states linked by four relations. PhyloVelo's displayed value of +1.000 is an adapted diagnostic: repeating its global gene-wise vector across cells makes within-state coherence structurally one. It is excluded from comparable GDC ranking and does not establish superior performance. The other 19 scores are unchanged. TopicVelo uses a native-transition displacement proxy.